IGCSE 0607 Practice Questions — E8.3

Magnitude of a vector

Total Questions: 15 — **Total Marks:** 100 Marks

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1 Vector $\mathbf{a} = \begin{pmatrix} -12 \\ 5 \end{pmatrix}$.

Calculate $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

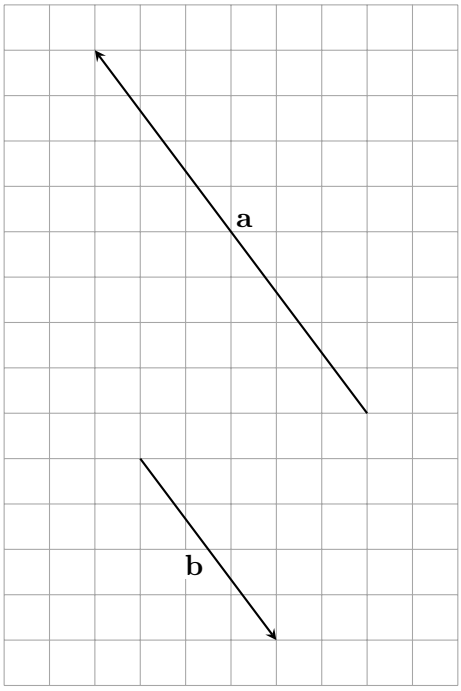
2 Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

3 The diagram shows vectors **a** and **b** drawn on a square grid.



(a) Work out $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

(b) Work out $|2\mathbf{a} + 3\mathbf{b}|$.

$|2\mathbf{a} + 3\mathbf{b}| =$ ..... **[3]**

4 Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

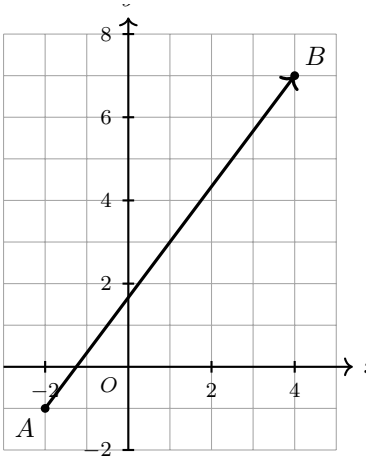
(a) Calculate $|\mathbf{p}|$.

..... **[2]**

(b) Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

5 The diagram shows points $A(-2, -1)$ and $B(4, 7)$ on a coordinate grid.



(a) Write down $\vec{AB}$ as a column vector.

$$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[1]**

(b) Calculate $|\vec{AB}|$.

$|\vec{AB}| =$ ..... units **[2]**

(c) The vector $\mathbf{v}$ is given by $\mathbf{v} = \begin{pmatrix} -12 \\ -5 \end{pmatrix}$.

Calculate

(i) $|\mathbf{v}|$,

$|\mathbf{v}| =$ ..... units **[2]**

(ii) $|\vec{AB} + \mathbf{v}|$, giving your answer in the form $k\sqrt{5}$, where k is an integer.

$|\vec{AB} + \mathbf{v}| =$ ..... units **[2]**

6 **(a)** The vector $\mathbf{a} = \begin{pmatrix} p \\ p \end{pmatrix}$, where $p > 0$.

Given that $|\mathbf{a}| = \sqrt{18}$, find the value of p.

$p =$ ..... **[2]**

(b) The vectors $\mathbf{u}$ and $\mathbf{v}$ are given by

$$\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

A vector $\mathbf{w}$ is defined as $\mathbf{w} = \mathbf{u} + t\mathbf{v}$.

Given that $|\mathbf{w}| = \sqrt{29}$, find the two possible values of t.

$t =$ ..... , or $t =$ ..... **[5]**

7 Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

(a) The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

[3]

(b) Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

(c) Hence write down the two possible column vectors for **u**.

[2]

8 Relative to an origin O, the position vectors of points A and B are $\vec{OA} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ and $\vec{OB} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

(a) Calculate the exact value of

(i) $|\vec{OA}|$, giving your answer in the form $a\sqrt{10}$, where a is an integer,

$|\vec{OA}| =$ ..... **[1]**

(ii) $|\vec{AB}|$, giving your answer in the form $b\sqrt{5}$, where b is an integer.

$|\vec{AB}| =$ ..... **[2]**

(b) Show that triangle OAB is both isosceles and right-angled.

[3]

(c) Point P lies on the line segment AB such that $AP : PB = 3 : 1$.

(i) Find $\vec{OP}$ as a column vector.

$\vec{OP} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(ii) Point Q has position vector $\vec{OQ} = \begin{pmatrix} 8 \\ m \end{pmatrix}$, where $m < 0$.

Given that $|\vec{PQ}| = 17$, find the value of m.

$m =$ ..... **[3]**

9 Points P and Q have coordinates $(4, -2)$ and $(-1, 10)$ respectively.

(a) Find $\vec{PQ}$.

$$\vec{PQ} = \begin{pmatrix} \quad \\ \quad \end{pmatrix} \quad \textbf{[1]}$$

(b) Calculate $|\vec{PQ}|$.

$|\vec{PQ}| =$ ..... **[2]**

(c) Vector $\mathbf{m} = \begin{pmatrix} 3 \\ c \end{pmatrix}$.

Given that $|\mathbf{m}| = |\vec{PQ}|$, find the possible values of c.

$c =$ ..... , $c =$ ..... **[3]**

10 Points A, B, and C have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

(a) Calculate $|\vec{AC}|$.

..... **[2]**

(b) The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of y.

$y =$ ..... , $y =$ ..... **[5]**

11 The points P, Q and R have coordinates $(2, -3)$, $(k, 5)$ and $(4, 1)$ respectively.

(a) Given that $|\vec{PQ}| = 10$, find the two possible values of k.

$k =$ ..... , or $k =$ ..... **[4]**

(b) Using the positive value of k from part (a), calculate the magnitude of $\vec{QR}$.
Give your answer correct to 3 significant figures.

$|\vec{QR}| =$ ..... **[3]**

12 **(a)** The points A and B have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

(c) Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

13 $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$.

(a) Calculate $|2\mathbf{p} - \mathbf{q}|$, giving your answer correct to 3 significant figures.

$|2\mathbf{p} - \mathbf{q}| =$ ..... **[3]**

(b) The vector $\mathbf{r} = k\mathbf{p}$, where $k < 0$ and $|\mathbf{r}| = 35$.
Find the column vector $\mathbf{r}$.

$\mathbf{r} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

(c) The vector $\mathbf{w} = \begin{pmatrix} t+3 \\ 2t-1 \end{pmatrix}$ has magnitude $\sqrt{34}$.
Find the two possible values of t.

$t =$ ..... , $t =$ ..... **[4]**

14 Vector $\mathbf{v} = \begin{pmatrix} k+3 \\ 2k-1 \end{pmatrix}$, where k is a constant.

The magnitude of $\mathbf{v}$ is $\sqrt{98}$.

(a) Show that $5k^2 + 2k - 88 = 0$.

[3]

(b) Solve the equation $5k^2 + 2k - 88 = 0$ to find the possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

(c) Given that $k > 0$, write down the column vector $\mathbf{v}$.

$\mathbf{v} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

15 Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

(a) Given that $|\mathbf{u} + \mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

[4]

(b) Solve the equation

$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

(c) Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**